# Finolex Academy of Management and Technology, Ratnagiri

Department of Computer Science and Engineering(AI & ML)

A Project Presentation on

# OWN HR SOLUTION 2.0 – Intelligent Talent and Resume Management


By,

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|---|---|
| Janhavi Deepak Bandbe | R220093 |
| Vinit Vinod Kedari | R220136 |
| Shravita Pravin Kelkar | R220544 |
| Sushant Bhanudas Kocharekar | R220078 |

**Guided By - M. M. Hatiskar**

# Introduction

- **Background-**

The existing HR management system faces challenges in terms of outdated architecture, inefficient resume matching, and limited scalability. With the growing demand for automation and personalized job recommendations, there is a need for a more robust and intelligent system.

- **Purpose-**

To redevelop the current HR solution into a modern, efficient, and scalable platform with improved features like smart resume matching, secure authentication, optimized data handling, and an intuitive user interface.

# How
# OWN HR SOLUTIONS WORK ?

1

**Candidates submit**

Candidates submit resumes to Own HR Solution through the platform, either directly or via **employees/freelancers.**

2

**AI-Powered Matching**

Resumes through the platform. The system uses AI to extract key details (skills, experience, education) and creates a digital profile, which is stored in the database for matching with client job openings.

3

**Revenue Model**

When a candidate joins the client company, Own HR Solution receives a payout (payment) from the client.

Payment is usually per hire and can be either a fixed fee or a percentage of the candidate’s salary.

# Objectives

- **Primary Objectives-**

To design and develop a next-generation HR management platform that enhances recruitment operations and user experience.

- **Secondary Objectives-**

- Redesigning UI/UX for better accessibility and performance
- Integrate AI-driven resume parsing and matching
- Ensure Mobile Responsiveness
- Resume analyzing and updating

# Literature Review

| Sr. No. | Paper Title | Author(s) | Publication Details | Identified Gaps | Proposed Solutions (Our Project) |
|---|---|---|---|---|---|
| 1 | Application of LLM Agents in Recruitment: A Novel Framework for Resume Screening | Chengguang Gan, Qinghao Zhang, Tatsunori Mori | **Publisher:** arXiv / IPSJ **Published in:** 2024 | Focuses on theoretical LLM framework, lacks full-stack implementation | Build an **end-to-end MERN + BERT system** with recruiter dashboard |
| 2 | Intelligent Resume Matching System | Y. Sagar, K. S. G. Prakeerth, G. Abhiram, N. Mahideep, M. Poojith | **Publisher:** IJRASET **Published in:** 2025 | Embeddings-based matching but **no recruiter-friendly visualization** | Add **interactive React dashboard** with filters and ranking |
| 3 | Automated Resume Screening Tool | K. Aruna Kumari, M. Ramya, G. Mahesh, Y. Prasad, SK. Ashik | **Publisher:** IJITCE **Published in:** 2025 | Relies on MiniLM embeddings, lacks **robust preprocessing/scalability** | Use **BERT for deeper semantic analysis** + scalable **Node.js backend** |
| 4 | Resume Classification and Ranking using Transformer Models | J. Patel, K. Nair, H. Iyer | **Publisher:** ACM **Published in:** 2021 KDD Conference | Doesn't handle **multilingual resumes** or **regional needs** | Extend with **multilingual support (English, Hindi, Marathi)** |

# Project Scope

- **Inclusions –**

- Resume upload, parsing, and intelligent matching
- Admin, employee dashboards
- Secured login and authorization mechanisms
- Performance-focused frontend with responsive design.
- AI-based Resume analysis
- Interview Scheduling

# Methodology

- **Flowchart**

Employees access the Resume Analyzer through a secure React dashboard to submit their resumes.

The Node.js + Express backend handles the uploads and performs text extraction.

A BERT model processes the extracted text to identify skills, keywords, and relevant insights.

The analyzed results are saved in a MySQL database for reliable storage.

Insights are then delivered back to the dashboard for employees to review and enhance their resumes

# Methodology

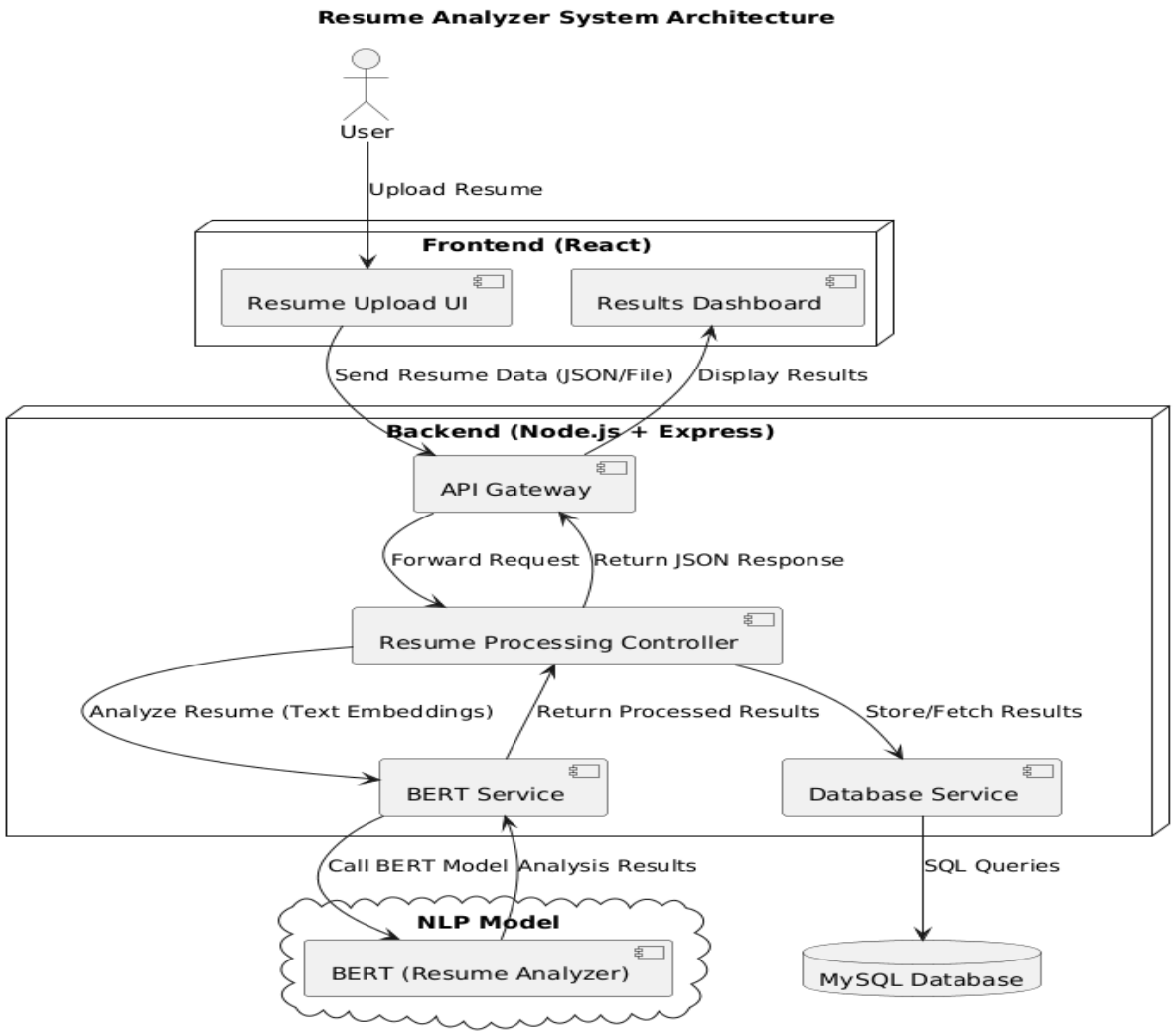
## ▪ UML Diagram

The Resume Analyzer System lets employees upload resumes via a **React frontend**.

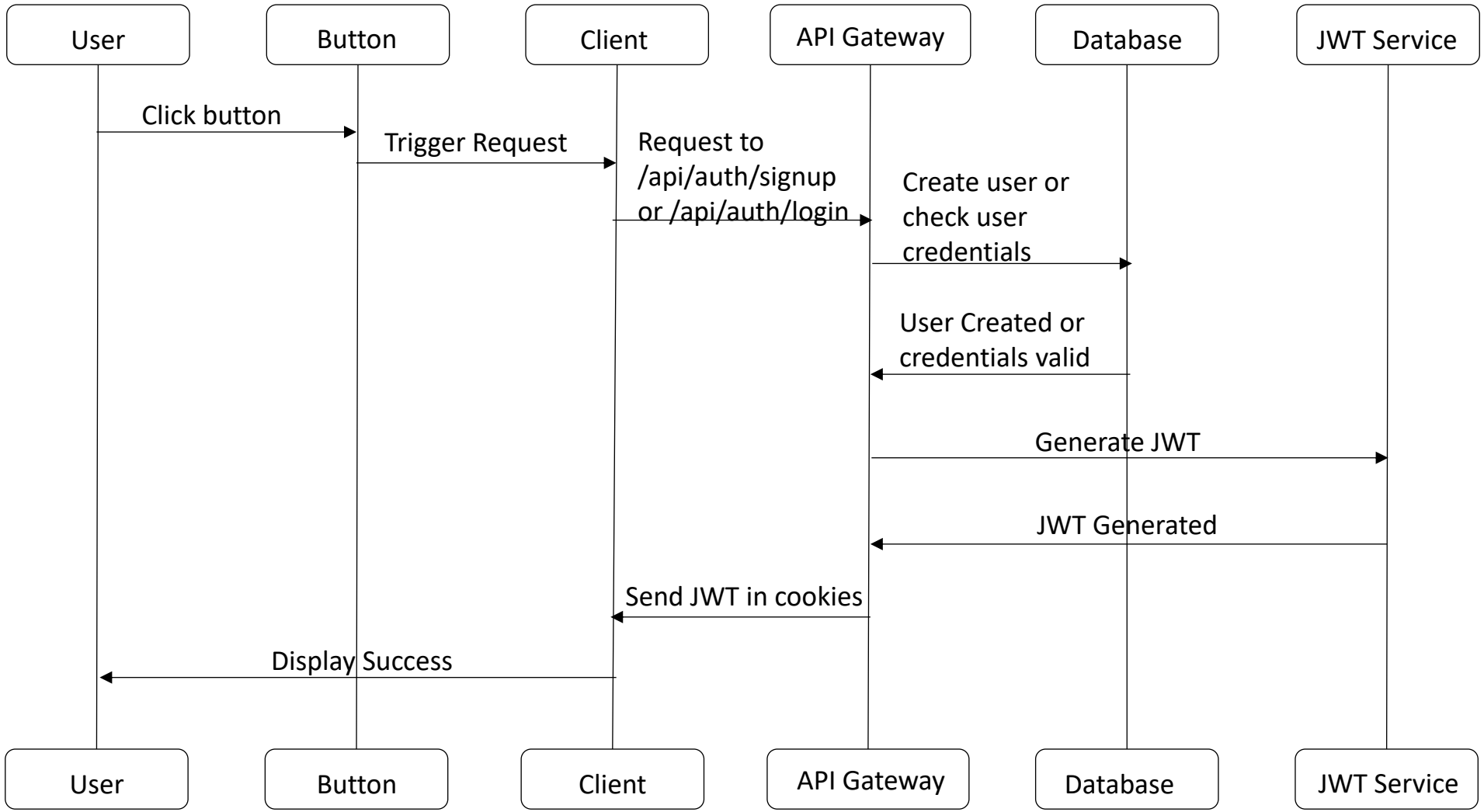
Requests are handled by a **Node.js + Express backend** with an API gateway and controllers.

Resumes are processed through the **BERT model** for skill and keyword analysis.

Results are stored and retrieved using a **MySQL database**.



# JSON Web Token(JWT) – Signup/Login



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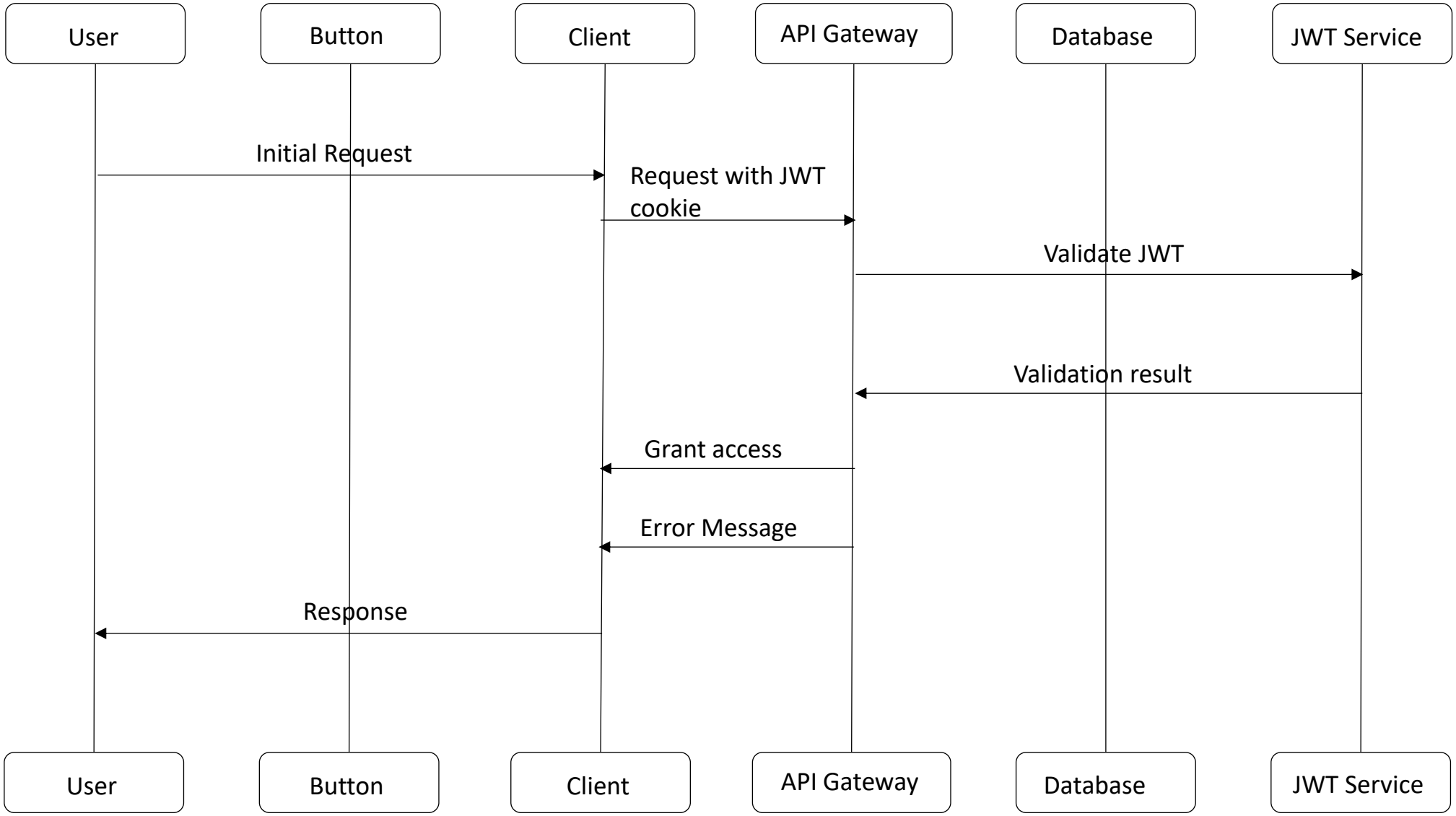
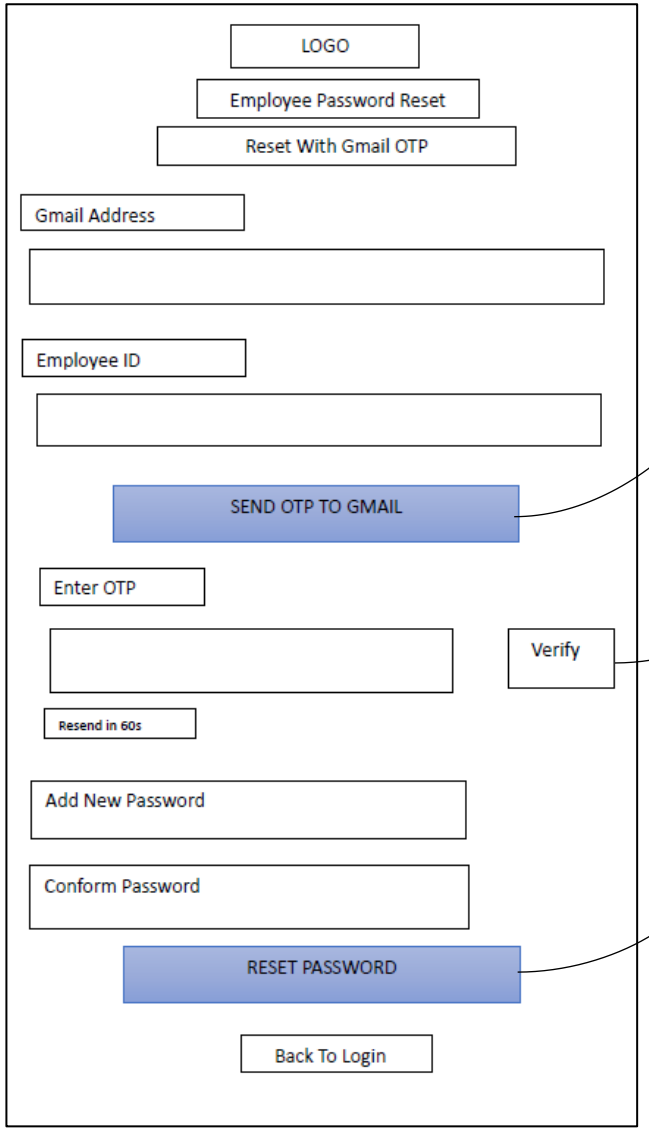
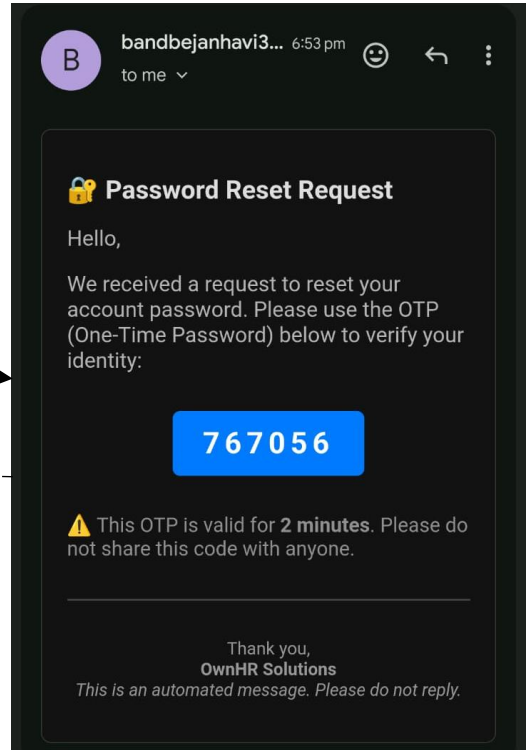


## Forgot Password




**Send OTP to Gmail**

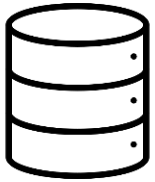
**Hashed OTP**




**Verify OTP**

**OTP verified = True**

**Reset Password**



# ▪ Activity Diagram

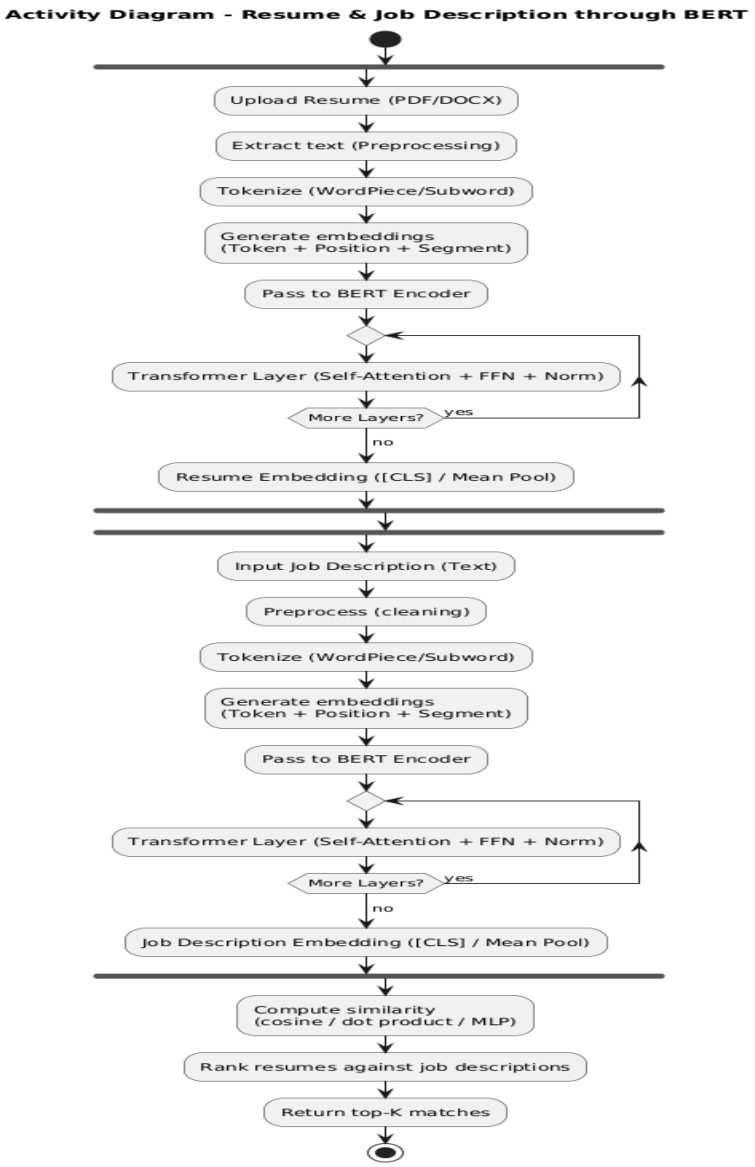
The system starts by **uploading a resume** (PDF/DOCX) and extracting text through preprocessing.

The text is **tokenized** and converted into embeddings, then passed through **BERT encoder layers** to generate a resume embedding.

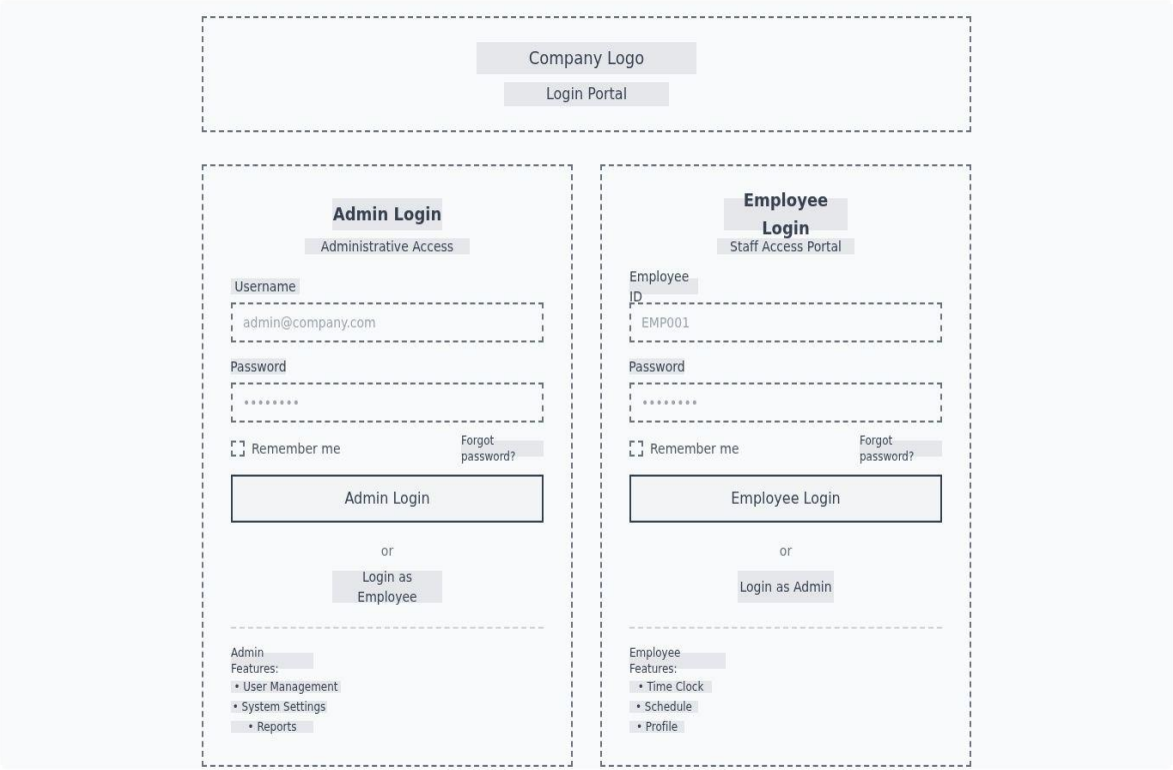
Similarly, the **job description text** undergoes preprocessing, tokenization, embedding generation, and BERT encoding.

Both resume and job description embeddings are obtained using **[CLS] token or mean pooling**.

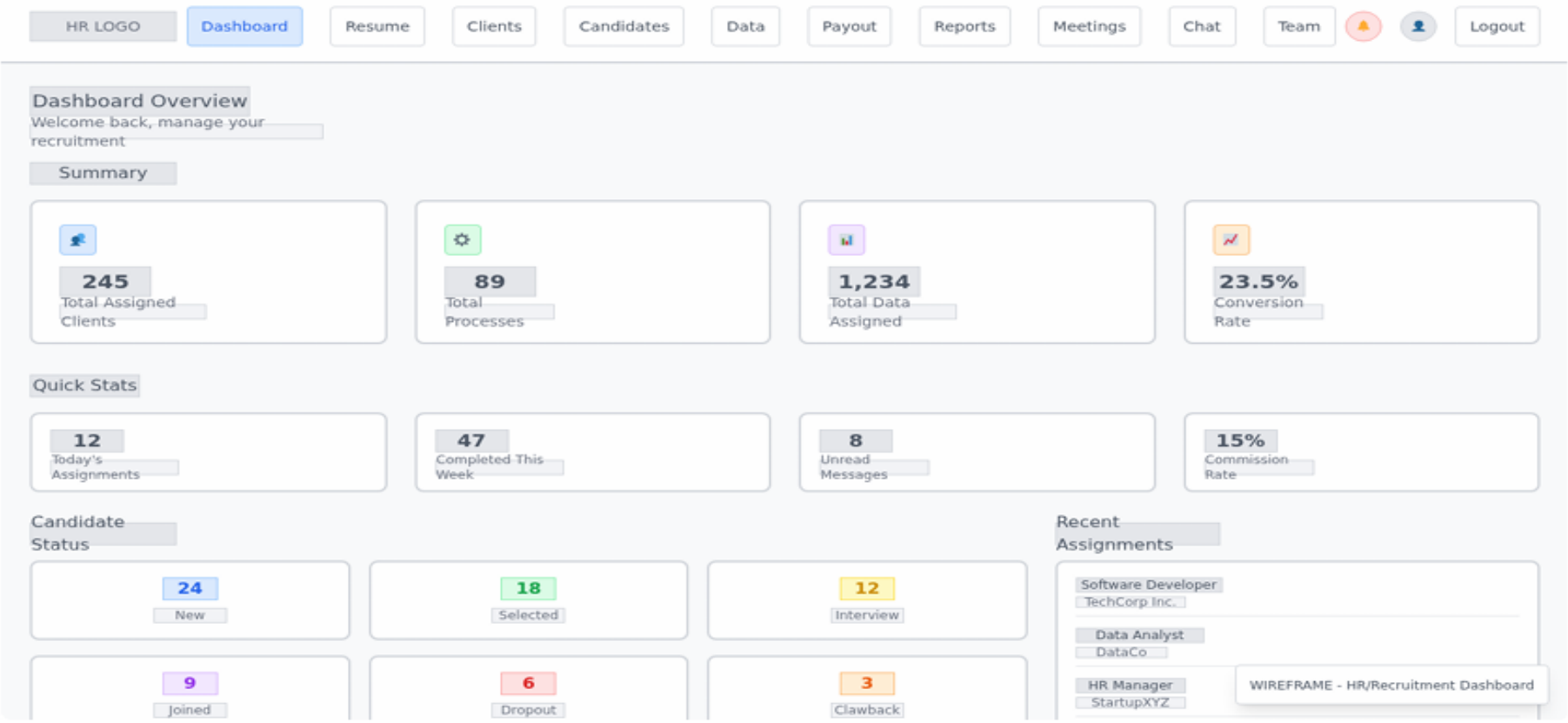
The system computes **similarity scores** (cosine, dot product, or MLP) between resumes and job descriptions. Finally, resumes are **ranked**, and the **top-K matches** are returned.



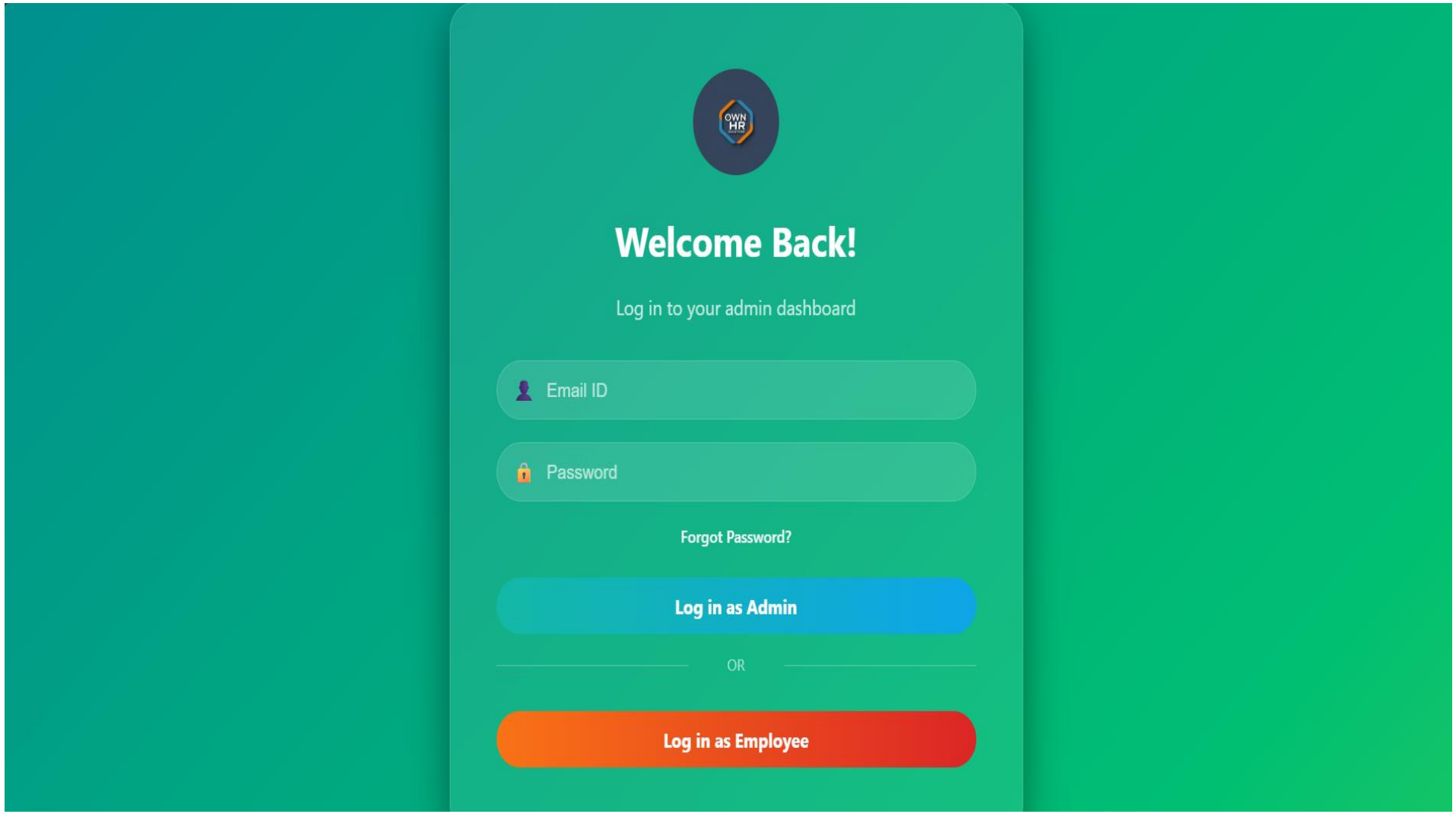
# Wireframe


Company Logo
Login Portal
Admin Login
Administrative Access
Username
admin@company.com
Password
••••••••
Remember me
Forgot password?
Admin Login
or
Login as Employee
Admin Features:
• User Management
• System Settings
• Reports
Employee Login
Staff Access Portal
Employee ID
EMP001
Password
••••••••
Remember me
Forgot password?
Employee Login
or
Login as Admin
Employee Features:
• Time Clock
• Schedule
• Profile

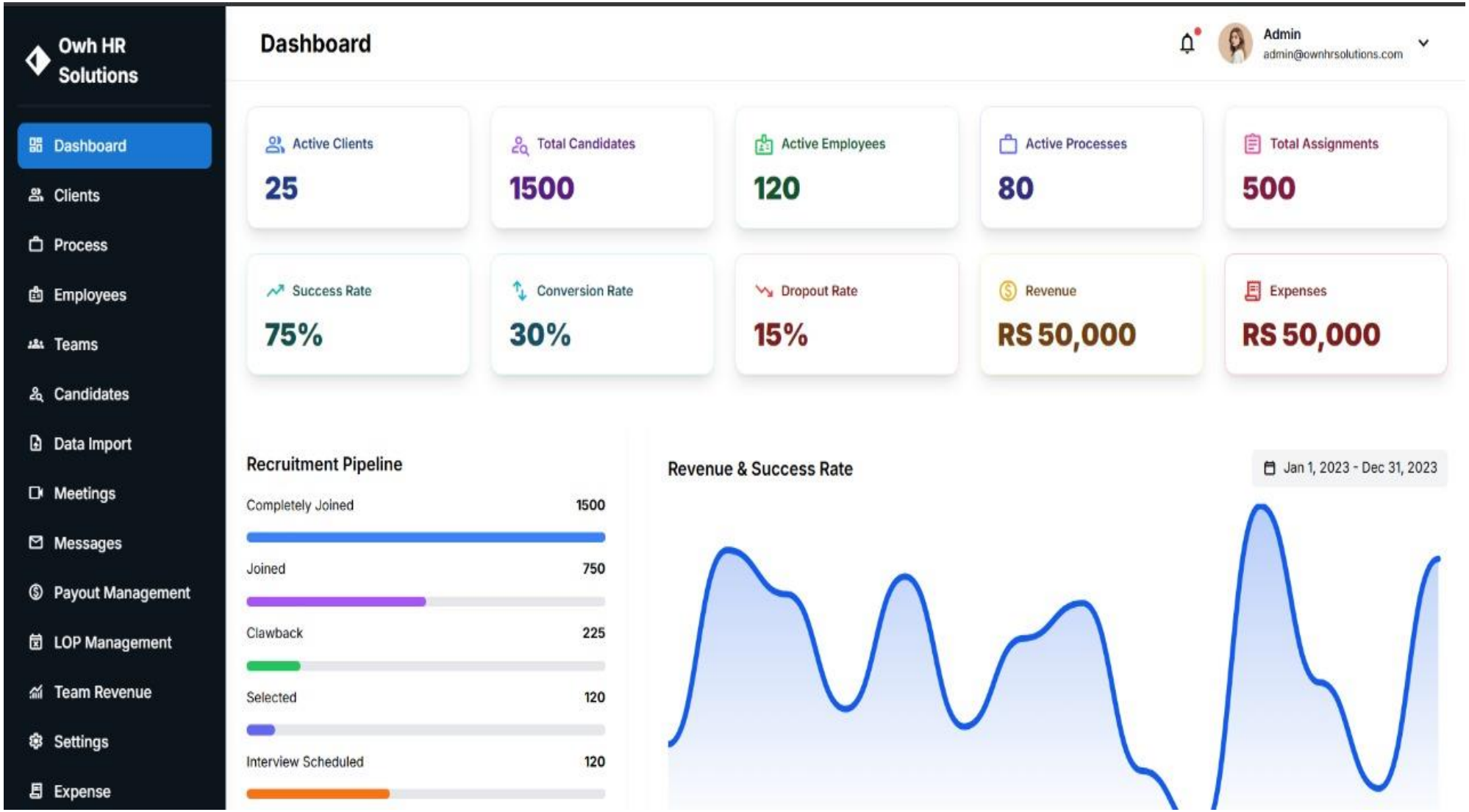
# Wireframe


HR LOGO
Dashboard
Resume
Clients
Candidates
Data
Payout
Reports
Meetings
Chat
Team
Logout
Dashboard Overview
Welcome back, manage your recruitment
Summary
245
Total Assigned Clients
89
Total Processes
1,234
Total Data Assigned
23.5%
Conversion Rate
Quick Stats
12
Today's Assignments
47
Completed This Week
8
Unread Messages
15%
Commission Rate
Candidate Status
24
New
18
Selected
12
Interview
9
Joined
6
Dropout
3
Clawback
Recent Assignments
Software Developer
TechCorp Inc.
Data Analyst
DataCo
HR Manager
StartupXYZ
WIREFRAME - HR/Recruitment Dashboard

# Implementation



# Implementation



# Conclusion

The redevelopment of **OWN HR SOLUTIONS** aims to bridge critical system gaps through AI-driven resume matching, secure authentication, modern and responsive interface. This project not only enhance recruitment efficiency and scalability but also align with academic objectives, serving as practical and innovatively.

Thank you